

Jaeyong Shin

Ph.D. Student in Robotics | Seoul National University
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[Google Scholar](#) [LinkedIn](#)

Research Interests

Reinforcement Learning, Sim-to-Real Transfer, Humanoid Robot, Bipedal Walking, and Whole-body control.

Education

Integrated M.S./Ph.D. Program

Seoul, Korea

Department of Convergence Science and Technology
DYROS Lab (Dynamic Robotics & Systems Lab)
Formerly supervised by Prof. Jaeheung Park (2026)
GPA: 4.01/4.30

Seoul National University

Sep 2021–Present

B.S. in Mechanical Engineering

Seoul, Korea

Yonsei University

Mar 2015–Aug 2021

Publications

Journal Articles

- [J4] **Jaeyong Shin***, Woohyun Cha*, and Jaeheung Park, “Learning Footstep Constrained Policies Guided by Linear Inverted Pendulum Model for Humanoid Robots,” *IEEE Access* (under review after major revision, 2026). *Equal contribution (co-first authors), [supplementary video](#)
- [J3] Eunho Sung, Seungbin You, **Jaeyong Shin**, ..., Jaeheung Park, “SNU-Avatar Haptic Arm: A Hybrid QDD/DD Operator Interface for Teleoperation,” *IEEE Access* (in press)
- [J2] Beomyeong Park, Donghyeon Kim, Daegyu Lim, ..., **Jaeyong Shin**, et al., “Intuitive and Interactive Robotic Avatar System for Tele-Existence: Team SNU in the ANA Avatar XPRIZE Finals,” *International Journal of Social Robotics*, vol. 17, no. 3, pp. 379–407, 2025. [\[DOI\]](#)
- [J1] Myeong-Ju Kim, Daegyu Lim, Donghyeon Kim, Junhyeok Cha, **Jaeyong Shin**, Woohyun Cha, Gyeongjae Park, Kwanwoo Lee, and Jaeheung Park, “Advances in Humanoid Robot Walking Technologies: A Review,” *Journal of Institute of Control, Robotics and Systems*, vol. 30, no. 4, pp. 412–422, 2024. [\[DOI\]](#)

Conference Papers

- [C4] **Jaeyong Shin**, Junewhee Ahn, Suhan Park, et al., “Virtual Reality Based Intuitive Spatial Visual Interface for Avatar Robot System,” *23rd Int’l Conf. on Control, Automation and Systems (ICCAS)*, 2023. [\[DOI\]](#)
- [C3] Hokyun Lee, Gyeongjae Park, **Jaeyong Shin**, Beomyeong Park, and Jaeheung Park, “Foot-Operated Telelocomotion Interface for Avatar Robots Utilizing Mecanum Wheel-Based Mobile Platforms,” *23rd Int’l Conf. on Control, Automation and Systems (ICCAS)*, 2023. [\[DOI\]](#)
- [C2] Junhyung Kim, Beomyeong Park, **Jaeyong Shin**, and Jaeheung Park, “Analysis of Effects of Leg Inertia Distribution on Contact Stability and Dynamic Performance During Bipedal Walking,” *IEEE RAS 21st Int’l Conf. on Humanoid Robots (Humanoids)*, 2022. [\[DOI\]](#)
- [C1] **Jaeyong Shin**, Junewhee Ahn, and Jaeheung Park, “Stereoscopic Low-Latency Vision System via Ethernet Network for Humanoid Teleoperation,” *19th Int’l Conf. on Ubiquitous Robots (UR)*, 2022. [\[DOI\]](#)

Preprints

- [P2] **Jaeyong Shin**, Woohyun Cha, Donghyeon Kim, Junhyeok Cha, and Jaeheung Park, “Spectral Normalization for Lipschitz-Constrained Policies on Learning Humanoid Locomotion,” *arXiv:2504.08246*, 2025. [arXiv]
- [P1] Junhyeok Cha, Woohyun Cha, **Jaeyong Shin**, Donghyeon Kim, and Jaeheung Park, “Sim-to-Real of Humanoid Locomotion Policies via Joint Torque Space Perturbation Injection,” *arXiv:2504.06585*, 2025. [arXiv]

Research Projects

Foot-Constrained Bipedal Locomotion (2D & 3D)	2025–Present
Developed a Foot-constrained RL bipedal walking controller using FIPM/LIPM with VRP/ZMP and preview control for CoM trajectory generation, enabling stair climbing and unstructured terrain traversal. Platform: Unitree G1, TOCABI. <i>Manuscript in preparation</i> — target venue: <i>IEEE RA-L</i> <i>Experimental Videos</i> — https://youtu.be/6c7JaemRq60 , https://youtu.be/CbqO_JDeBl8	
Spectral Normalization for Lipschitz-Constrained Policies	2025–Present
Proposed Spectral Normalization (SN) as a memory-efficient alternative to gradient-penalty-based Lipschitz-Constrained Policies (LCP), suppressing high-frequency torque fluctuations from unrealistic simulation assumptions while achieving comparable performance with significantly reduced GPU memory overhead. <i>Manuscript in Arxiv</i> — target venue: <i>ICRA after follow-up study</i> <i>Experimental Videos</i> — https://youtu.be/Kr26Y5CayyU	

Projects

인간 수준의 다중 감각 센싱 시스템과 정밀 조작 원격 인터페이스를 갖춘 인간형 아바타 로봇 개발	2024–2025
Developed a long-distance remote teleoperation interface linking Stanford University and Seoul National University.	
열악 환경 비정형 작업을 위한 3D협업 텔레오퍼레이션 기술 개발	2023–2025
Developed an operator teleoperation interface system for intuitive avatar robot control.	
내재 센서 기반 전신 보행 제어 및 실시간 동작 계획	2023–2024
Developed a reinforcement learning-based torque-level bipedal walking policy	
ANA Avatar XPRIZE — Team SNU	2021–2022
Developed stereoscopic low-latency Ethernet vision and VR-based spatial interface for TOCABI humanoid teleoperation. Finalist in the \$10M ANA Avatar XPRIZE global competition (top teams world-wide).	

Affiliations

Graduate Researcher, DYROS Lab, Seoul National University	2021–Present
Software Project Lead, Bluerobin co.	Sep–Dec 2025
<i>Humanoid Robot P-73 Project</i> — sole architect of full middleware stack and real-time locomotion controllers	

Technical Skills

Robotics	Reinforcement Learning, Sim-to-Real Transfer, Bipedal locomotion, Whole-body control.
Platforms	Unitree G1, TOCABI, P-73
Programming	Python, C++, ROS

Languages

Korean (Native)	English (Fluent)
TOEFL iBT: 101	(R27 / L26 / S22 / W26, Jun 2019)